

Nguyen Hieu Kha

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About Me

AI engineer with a strong foundation in Natural Language Processing (NLP), Computer Vision (CV), Generative AI (GenAI). I am seeking an opportunity to apply my knowledge to real-world projects.

Education

Saigon International University, BS in Artificial Intelligence Sep 2022 – Apr 2026

- GPA: 3.6/4.0, full 4-year Scholarship (100% tuition fee).

Experience

AI Intern, Gon Tech Asia Mar 2026 – Jun 2026

- Built web search and RAG services for a Multi-Agent System, with multi-layer caching to reduce repeated planning, retrieval, and response generation.
- Implemented prompt caching across agents and LLM providers, using Langfuse to trace latency, token usage, and cache hits, based on benchmarks showing **68.5% LLM cost reduction**.

AI Intern, Vietnam Communication Corporation — VCCorp Oct 2025 – Feb 2026

- Built an automated data pipeline for dataset construction and quality validation, reducing **manual data costs by 93.3%** and **manual annotation time by 80%**.
- Reproduced state-of-the-art sign language translation models to benchmark dataset quality and analyze data statistics.

AI Intern, The Data and Artificial Intelligence Center — SIU AI Lab Jun 2024 – Oct 2024

- Collaborated on R&D for a cancer-support medical chatbot, focusing on contextual response quality and patient-centered experience.
- Explored advanced RAG and multi-agent techniques to improve retrieval quality and response reliability.

Projects

Audio-based LLM: Vietnamese Real-time Voice Chat System  GitHub Repo

- Built a real-time Vietnamese voice chat system with a browser client and FastAPI WebSocket backend, integrating Whisper STT, Gemini LLM, and a fine-tuned Piper TTS model.
- Achieved real-time performance with average **RTF 0.46** and **latency about 1.7 s**.

Temporal Coherence Framework for Multimodal Video Retrieval  GitHub Repo

- Built a video event retrieval system using multimodal fusion and temporal modeling for efficient text- and image-based search.
- Integrated CLIP-based retrieval, PaddleOCR, Whisper for speech recognition, and AutoShot for shot segmentation to improve semantic and temporal understanding of video content.
- Evaluated the system on the AIC2025 dataset, achieving **95% accuracy** with **query latency under 1 s**.

ViSL Tool: Vietnamese Sign Language Dataset Builder  GitHub Repo

- Built ViSL Tool, a full-stack system using FastAPI, PostgreSQL, Next.js, and Docker for automated collection and annotation of Vietnamese Sign Language data, creating one of the first sentence-level VSL datasets in Vietnam.
- Developed a data pipeline with yt-dlp, OpenCV, WhisperX, and Python for video collection, filtering, transcription, and annotation generation.
- Evaluated the pipeline with **99.52% accuracy** in video filtering and reduced text normalization **CER from 6.0% to 3.2%**.

Honors & Awards

Potential Prize — AI Challenge 2025 Nov 2025

- Awarded for the project: “Cross Segment Coherence Scorer: A Training Free Temporal Framework for Multimodal Video Retrieval.”

Consolation Prize — 26th Euréka Student Research Award 2024 Jan 2025

- Awarded for the project: “Developing a Chatbot to Support the Assessment of Mental Health Conditions in University Students.”

Consolation Prize — AI Challenge 2024

Nov 2024

- Awarded for the project: “Enhanced Video Event Retrieval Using Fusion and Temporal Modeling.”

First Prize, SIU Student Research Competition

Apr 2024

- Awarded for the project: “Developing a Chatbot to Support Mental Health Assessment for Students at Saigon International University.”

Skills

Languages: English (IELTS 6.5 – Professional Working Proficiency), Vietnamese (Native)

Programming & Frameworks: Python, C, C++, Java, PyTorch, TensorFlow, LlamaIndex, LangChain.

Web & Tools: JavaScript, Flask, FastAPI, Faiss, Qdrant, Milvus, Git, GitHub, GitLab CI/CD, Docker.

Publications

Conference Proceedings

Apr 2025

- H. T. Quang, H. L. Pham, **H. K. Nguyen**, *et al.*, “Vizquest: Enhanced video event retrieval using fusion and temporal modeling,” in *Information and Communication Technology*, W. Buntine, M. Fjeld, T. Tran, *et al.*, Eds. Singapore: Springer Nature Singapore, 2025, pp. 155–166. ISBN: 978-981-96-4291-5. doi:10.1007/978-981-96-4291-5_13
- **H. K. Nguyen**, D. L. Phat, P. N. Duc, *et al.*, “Cross-Segment Coherence Scorer: A Training-Free Temporal Framework for Multimodal Video Retrieval,” accepted at *SoICT 2025*, to appear in conference proceedings.